

Scope for Quality Audit of Wagon Depot:

1. Quality audit of wagon depot for ROH attention of freight stock (Air brake stock fitted with CASNUB bogies and BLC wagon).
2. Quality audit of wagon depot for Yard attention (Intensive examination) and sick-line attention.
3. Quality audit of Wagon Depot with regard to maintenance of Stainless Steel Wagon (BOXNHL / BCNHL/BRN22.9)

Quality Assurance Directorate (Mechanical)
RDSO Lucknow

**Quality audit check-sheet for ROH attention of
Freight Stock (Air brake stock fitted with CASNUB bogies and BLC wagon)**

(Based on CAMTECH Maintenance Manual for Wagon and CAMTECH report on Standardisation of Infrastructure Facilities for Freight stock maintenance)

SN	Requirements	Clause	Observations																																				
1	ROH ATTENTION																																						
1.1	ROH periodicity for various types of common wagon as stipulated is being followed: <table><tr><th>SN</th><th>Type of stock</th><th>Periodicity of ROH (Months)</th></tr><tr><td>1</td><td>Stainless steel wagon (BOXNCR/BONLW/BOXNHL)</td><td>18 month (for new stock BOXNHL – 1ST ROH - 24 months)</td></tr><tr><td>2</td><td>Other Open Wagon</td><td>18</td></tr><tr><td>3</td><td>Covered wagon</td><td>24</td></tr><tr><td>4</td><td>Hopper wagon</td><td>24</td></tr><tr><td>5</td><td>BOY wagon (22.9)</td><td>18</td></tr><tr><td>6</td><td>BOBS wagons</td><td>18</td></tr><tr><td>7</td><td>BOBYN</td><td>24</td></tr><tr><td>8</td><td>Bogie tank wagon (BTPN)</td><td>18</td></tr><tr><td>9</td><td>BTPGLN</td><td>24</td></tr><tr><td>10</td><td>BVZI</td><td>12</td></tr><tr><td>11</td><td>BLC</td><td>24</td></tr></table>	SN	Type of stock	Periodicity of ROH (Months)	1	Stainless steel wagon (BOXNCR/BONLW/BOXNHL)	18 month (for new stock BOXNHL – 1 ST ROH - 24 months)	2	Other Open Wagon	18	3	Covered wagon	24	4	Hopper wagon	24	5	BOY wagon (22.9)	18	6	BOBS wagons	18	7	BOBYN	24	8	Bogie tank wagon (BTPN)	18	9	BTPGLN	24	10	BVZI	12	11	BLC	24	Table 2.3 of wagon manual, ED committee report and RB's guideline for ROH of BOXNHL (2007/ M(N)/ 951 / 45)	
SN	Type of stock	Periodicity of ROH (Months)																																					
1	Stainless steel wagon (BOXNCR/BONLW/BOXNHL)	18 month (for new stock BOXNHL – 1 ST ROH - 24 months)																																					
2	Other Open Wagon	18																																					
3	Covered wagon	24																																					
4	Hopper wagon	24																																					
5	BOY wagon (22.9)	18																																					
6	BOBS wagons	18																																					
7	BOBYN	24																																					
8	Bogie tank wagon (BTPN)	18																																					
9	BTPGLN	24																																					
10	BVZI	12																																					
11	BLC	24																																					
1.2	ROH OF AIR BRAKE WAGON WITH CASNUB BOGIE	Para 205 (Chapter 2) of CAMTECH Wagon manual.																																					
1.2.1	a) Lift the body, keep it on trestles and run out bogies b) Strip bogie component for examination and repair as below: ➤ Strip spring & spring suspension arrangement including snubbing device. ➤ Check springs for free height and other defects. Replace where required. ➤ Examine Bogie frame. Check frame alignment as per instructions contained in RDSO Technical Pamphlet No. G-64. ➤ Examine pivot for welding defects/cracks/ abnormal depth due to wear. Replace where necessary and lubricate with graphite flakes to IS:495 in dry condition. c) Strip brake gear levers and rods for examination of worn out/damaged parts. d) The equipment shall be given attention in accordance with the maintenance manual issued by respective air brake equipment manufacturer:- ➤ Cleaning of strainer discs (Reference 6 & 218 of the pamphlet SBA 795/258 of P4aG type Distributor Valve) ➤ Lubrication of brake cylinder/cleaning of its strainer ➤ Check for easy operation of isolating cock & anti-pilferage device of distributor valve cut-off angle cock , <td></td> <td></td>																																						

SN	Requirements	Clause	Observations
	manual quick release valve & isolating cock. ➤ Draining of Auxiliary Reservoir ➤ Checking of hose coupling for serviceability ➤ Cleaning of strainer ➤ Dirt collector to be cleaned ➤ Leakage in pipes and joints to be checked ➤ After carrying out above items of work the wagon shall be tested for proper function of air brake system with single car test device in accordance with the procedure indicated below : ✓ Connect the BP coupling of single car test rig to the corresponding coupling of the wagon to be tested .The couplings of the other end of the wagon to be closed with dummy coupling heads .Fix pressure gauge on the brake cylinder. ✓ The single car test device should now be coupled to the main line of a compressor through a pipe. ✓ Carry out the preparation and testing in accordance with the procedure given in the manufacturers Maintenance Manual and record the reading in the test proforma. ✓ For passing the wagon, all the parameters shall be within specified limits e) Clean horizontal lever, hand brake & gears and lubricate f) Examine head stock for damage, bent/cracks g) Refit brake gear levers and rods, lubricate pins and other equipment and apply graphite to horizontal levers of empty load box h) Replace worn out brake blocks i) Check wheel profile, turn the wheels as needed. UST of Axle to be carried out & turning of wheel to worn wheel profile during ROH j) All the wheels are to be checked ultrasonically and Axle box cap bolts are to be tightened up by torque wrench with proper torque and in no case old locking plates are to be reused. k) CBC Knuckles are to be checked by contour gauge, anti creep/ articulated rotary operation of locking arrangement to be checked. l) Manual adjustment of brake gear to be done in accordance with wheel diameter. m) Modification works are to be attended as issued by RDSO from time to time.		
1.2.2	During ROH attention, schedule Items as per table 2.4 of CAMTECH manual for wagons are attended as under a) BODY REPAIR WORK (Common for all) ➤ UNDERFRAME- Sole Bar ✓ Scrap the portion of sole bar at doorways, clean and apply primer paint followed by Top coat ➤ SIDE WALL- Skirting ✓ Check and patch if corroded then apply primer and top coat on the patch ✓ Side Doors- Check damage and repair clean & lubricate hinges ✓ Side Pillars- Check cracks at the base & repair. ✓ Patching of body, roof, door or floor plates straightening bulged ends repairs to angle irons, stanchions and crib angles etc. ✓ Making wagons water-tight (Covered wagons) ✓ Fitting/replacement of door fastening road, door cotter pin eye, bolt hook and eye locking pin on empty wagons and closing of doors.	Table 2.4 (Chapter 2) of CAMTECH Wagon manual.	

SN	Requirements	Clause	Observations
	✓ Fitting/replacement of door fastening rod guide, door hasp which requires riveting on empty wagons. Repairs by welding to door fittings. ✓ Hand rail/foot board deficient/or damaged, insecurely fastened at door way of guard's brake van b) UNDER GEAR REPAIR WORK (Common for all) ➤ Brake Linkage- Check free movement ➤ Hand Brake - Check proper working c) BOGIE REPAIR WORKS (CASNUB BOGIE) ➤ BOLSTER ✓ Pocket slope liner Change liner if thickness less than 5 mm ✓ Rotation stop lugs Provide liners (thickness to suit) if dimensions less than 514 mm ✓ Inner Column Gib Provide liners (thickness to suit) if dimension more than 142 mm ✓ Land surface Provide liners (thickness to suit) if dimension less than 442 mm ✓ Outer Column Gib Renew by welding if dimension more than 241 mm ➤ SIDE FRAME ✓ Column Friction Liner - Change liner if dimension more than 455 mm ✓ Column sides - Provide liners (thickness to suit) if dimension less than 209 mm ✓ Anti rotation lugs - Provide liners (thickness to suit) if dimension more than 526 mm ✓ Key seat to pad 22 W - Provide liners (thickness to suit) if dimension more than 276 mm ✓ Crown Roof 22WM - Provide liners (thickness to suit) if dimension more than 321 mm ✓ Crown Roof 22NL - Provide liners (thickness to suit) if dimension more than 326 mm ✓ Pedestal Crown Sides - Renew by welding if dimension less than 147 mm ✓ Pedestal Jaw 22 W - Provide liners (thickness to suit) if dimension more than 275 mm ✓ Pedestal Jaw 22 WM - Provide liners (thickness to suit) if dimension more than 283 mm ✓ Pedestal Jaw 22 NL(Short) - Provide liners (thickness to suit) if dimension more than 195 mm ✓ Pedestal Jaw 22 NL(Long) - Provide liners (thickness to suit) if dimension more than 241 mm ✓ Pedestal Sides 22 W - Provide liners (thickness to suit) if dimension less than 102 mm ✓ Pedestal Sides 22 WM - Provide liners (thickness to suit) if dimension less than 102 mm ✓ Pedestal Side 22 NL - Provide liners (thickness to suit) if dimension less than 78 mm -- X X ➤ WEDGE ✓ Slope Surface - Renew by welding if dimension less than 7 mm ✓ Vertical Surface - If vertical surface from centre line of spigot less than 56 mm provide liner of 6 mm thickness		

SN	Requirements	Clause	Observations
	➤ CENTRE PIVOT (BOTTOM) ✓ Vertical Side 22 W - Renew by welding if wear more than 4 mm ✓ Vertical Side 22 WM - Renew by welding if wear more than 3 mm ✓ Vertical Side 22 NL - Renew by welding if wear more than 3 mm ✓ Seat Side 22 W Renew by welding if wear more than 3 mm ✓ Seat Side 22WM Renew by welding if wear more than 3 mm ✓ Seat Side 22 NL Renew by welding if wear more than 3 mm ➤ COIL SPRING ✓ Outer - Group and use in sets. Replace if free height at or less than 245 mm ✓ Inner - Group and use in sets. Replace if free height less than 247 mm ✓ Snubber - Group and use in sets. Replace if free height less than 279 mm ➤ ADAPTER ✓ Crown Surface Replace if worn 3.5 mm or more - Side lug Replace if wear more than 3 mm on either side ✓ Thrust shoulder Replace if depth exceeds 0.7 mm ✓ Machined relief Replace if depth less than 0.8 mm ➤ BOGIE BRAKE GEAR ✓ Pins & Bushes - Change if clearance more than 1.5 mm		
	d) AIR BRAKE SYSTEM ➤ Distributor Valve Overhauling ✓ Distributor Valve - Test on SWTR ✓ DV Isolating Cock - Examine operation ✓ DV Release valve - Examine operation ✓ DV Filter - Clean ➤ BRAKE CYLINDER ✓ Filter of Escorts & RPIL make - Clean - Brake Cylinder of Greysham & WSF make Lubricate ➤ CUT OFF ANGLE COCK ✓ Angle cock Examine and lubricate ➤ DIRT COLLECTOR ✓ Dirt Collector - Clean ✓ Sealing Ring - Change ➤ RESERVOIR ✓ AR & CR - Drain ✓ Sealing Ring - Change ➤ METAL PIPES & JOINTS ✓ Pipe Joints Examine leakage & repair ✓ Seals (20 mm & 32 mm) pipe - Change ✓ Guard's Emergency Brake Valve - Examine operation ✓ Isolating cock of BVZC Brake Van - Examine operation ✓ Quick Coupling - Examine operation ✓ Load Sensing Device - Examine operation		
	e) SLACK ADJUSTER (COMMON FOR ALL) ➤ Slack Adjuster - Test Functioning, repair if required ➤ "A" dimension - Adjust ➤ "e" dimension - Adjust ➤ M20 Anchor Pin nut - Ensure securing by welding to pin ➤ Air Brake System - Test on SWTR as per procedure		

SN	Requirements	Clause	Observations
	➤ Brake Block - Ensure Std. Key, Split pin & all new brake blocks f) CENTRE BUFFER COUPLER (COMMON FOR ALL) ➤ CBC BODY ✓ Coupler Body - Examine & replace on condition basis ✓ CBC Contour - Examine, replace if required ✓ Shank Wear Plate - Replace on condition ➤ KNUCKLE ✓ Nose - Replace if wear more than 9.5 mm with HTE knuckle ✓ Knuckle pin - Replace on condition ✓ Knuckle Stretch - Examine, Replace if required ➤ STRIKER CASTING ✓ Wear Plate - Replace ✓ Striker casting - Replace on condition ➤ COUPLER MECHANISM ✓ Anti Creep Protection - Examine and repair. ✓ Lock lift assembly – Examine ✓ Operation Mechanism - Examine ✓ Lock - Examine ➤ DRAFT GEAR ✓ Slack Measure & take correction ➤ GENERAL ✓ Yoke pin support - Replace on condition ✓ Buffer Height Examine & Correct if required g) WHEEL AXLE & BEARING (COMMON FOR ALL) ➤ AXLE ✓ Ultrasonic Testing - To be carried in every ROH & reject if fails ✓ Deep Notches due to gearing of pull rod - Reject if depth is more than 5 mm ✓ Axle end holes - Clean and lubricate in case end cover is opened ➤ WHEEL ✓ Tread profile - Check with tyre defect gauge ✓ Height of flange - Check with tyre defect gauge ✓ Smooth flange - If flange not completely smooth do not use under ROH wagon ✓ Wheel profile - Turn to WWP if above clause are not met for use under ROH wagon. ➤ Cartridge Tapered Roller Bearing ✓ Cup - Rotate the bearing for unusual sound check up for Crack/chipping ✓ Seal - Check seal for external damage/dent ✓ Backing Ring - Check backing ring for looseness & vent fitting on backing ring with vent hole (the vent fitting should be intact or the vent hole should be plugged) ✓ Locking Plate - Use new locking plate when ever and cover is opened ✓ Axle end Cap screw - Clean and lubricate in case end cover is opened ✓ Load Zone Change - Change load zone area of the cup while lowering bogie side frame		

SN	Requirements	Clause	Observations																
1.2.3	During ROH attention of bogie Nominal & Maximum Clearances in CASNUB Bogie are maintained as per table 2.4(Chapter 2) of CAMTECH wagon manual.	Table 2.4 (Chapter 2) of CAMTECH Wagon manual.																	
12.4	For BVZI wagon, bogies should be given IOH attention at depot. POH after 24 month to be done at workshop	RB letter no. 2003/M(N) / 951 / 17 dtd 30.03.11																	
1.3	ROH attention of BLC wagon	RITES maintenanc e manual for BLC wagon																	
1.3.1	After placement																		
a)	Visual checking of head stock for bent, crack as well as bogie maintenance																		
b)	Pre-testing of wagons: Test the wagon individually by SWTR to ensure the condition of DV, BC, angle cock as well as pipe line for any leakage																		
c)	Check the DV isolating cock, manual quick release valve																		
d)	Disconnect 'A' car from 'B' car – remove Yoke pin bolts, nuts with split pins and yoke pin support. If yoke pin is not dropping itself a bump through draw bar may enable the yoke pin to drop by its weight.																		
1.3.2	Disassembly of Bogies																		
a)	Disconnect the bogie brake rigging arrangement including rubber hose pipe and raise the wagon body after removing split pin, lock pin. Shackle lock etc. by using suitable over head cranes or 25 T whiting jack. Remove side frame key.																		
b)	Run out the bogies. Lower the under frame on the trestles. Remove the bogie by lifting with 4T capacity crane from the wheel sets. Place on fixture. Take out the bolster for repair.																		
1.3.3	After Lifting:																		
a)	Strip Springs & Springs Suspension arrangement. Check Springs for Free height and other defects. Replace when required. It is recommended that Springs having not more than 3mm Free Height variation should be assembled in same group.Mixing of New & Old Springs should be avoided. Springs should be marked as per grouping. <table border="1"><thead><tr><th colspan="3">Spring Free Height</th></tr><tr><th></th><th>Nom</th><th>Cond.</th></tr></thead><tbody><tr><td>Outer</td><td>260</td><td>245</td></tr><tr><td>Inner</td><td>243</td><td>228</td></tr><tr><td>Snubber</td><td>288</td><td>274</td></tr></tbody></table>	Spring Free Height				Nom	Cond.	Outer	260	245	Inner	243	228	Snubber	288	274			
Spring Free Height																			
	Nom	Cond.																	
Outer	260	245																	
Inner	243	228																	
Snubber	288	274																	
b)	Examine Bogie Frame, Frame Alignment by Dismantling the Bogie. Side frame to be checked by proper gauges to inspect wear & tear of friction liners. Worn out portion to be built up by proper welding in down hand position. Worn out liners to be replaced. <table border="1"><thead><tr><th>A.</th><th>BOGIE SIDE FRAME</th><th>Nominal</th><th>(Wear Limit)</th></tr></thead><tbody><tr><td>i)</td><td>Friction Wear Plate</td><td>10</td><td>(2+2)</td></tr><tr><td>ii)</td><td>Column Side</td><td>216</td><td>(5+5)</td></tr><tr><td>iii)</td><td>Anti Rotation Lug</td><td>450</td><td>(3+3)</td></tr></tbody></table>	A.	BOGIE SIDE FRAME	Nominal	(Wear Limit)	i)	Friction Wear Plate	10	(2+2)	ii)	Column Side	216	(5+5)	iii)	Anti Rotation Lug	450	(3+3)		
A.	BOGIE SIDE FRAME	Nominal	(Wear Limit)																
i)	Friction Wear Plate	10	(2+2)																
ii)	Column Side	216	(5+5)																
iii)	Anti Rotation Lug	450	(3+3)																

SN	Requirements	Clause	Observations																																																																																																
	<table><tr><th>B</th><th>PEDESTAL CROWN</th><th>Nominal</th><th>(Wear Limit)</th></tr><tr><td>i)</td><td>Key Seat to Pedestal</td><td>318</td><td>5</td></tr><tr><td>ii)</td><td>Pedestal Crown Side</td><td>152</td><td>4+4</td></tr><tr><td>iii)</td><td>Pedestal Side</td><td>105</td><td>2+2</td></tr><tr><td>iv)</td><td>Distance between Outer & Inner Pedestal Jaw</td><td>278</td><td>4+4</td></tr><tr><td>v)</td><td>Crown side</td><td>152</td><td>4+4</td></tr></table> <table><tr><th>C</th><th>ADAPTER</th><th>Nominal</th><th>(Wear Limit)</th></tr><tr><td>i)</td><td>Crown Lug</td><td>156</td><td>4+4</td></tr><tr><td>ii)</td><td>Side Lug</td><td>130</td><td>3+3</td></tr><tr><td>iii)</td><td>Adapter Side</td><td>268</td><td>3+3</td></tr></table> <table><tr><td>D.</td><td>BOGIE BOLSTER. Bolster pocket has 8 mm thick Si-Mg steel liner in slope pocket, may be permitted in service up to 3mm thick. Replace worn liner. No paint or grease should be applied on liner.</td><td>Nominal</td><td>(Wear Limit)</td></tr><tr><td>i)</td><td>Bolster Land Surface</td><td>443</td><td>3+3</td></tr><tr><td>ii)</td><td>Thickness of Slope Surface Liner</td><td>8</td><td>3+3</td></tr><tr><td>iii)</td><td>Rotation Stop Lug</td><td>518</td><td>3+3</td></tr><tr><td>iv)</td><td>Bolster Column Jib Outer</td><td>241</td><td>5+5</td></tr><tr><td>v)</td><td>Bolster Column Jib Inner</td><td>136</td><td>5+5</td></tr></table> <table><tr><th>E.</th><th>Center Pivot</th><th>(Wear Limit)</th></tr><tr><td>i).</td><td>Wear Limit Vertical side</td><td>2</td></tr><tr><td>ii)</td><td>Seat.</td><td>2</td></tr></table> <table><tr><th>F</th><th>Elastomeric Pad</th></tr><tr><td>i).</td><td>The top or the bottom plates should not show any cracks. If any sign of crushing of rubber is noticed reject the pad.</td></tr><tr><td>ii)</td><td>Permanent set in free height is allowed to 42 mm instead of 46 mm normal size. Use up-to 44 mm in ROH</td></tr></table> <table><tr><th>G</th><th>Side bearer</th><th>Nom</th><th>Cond</th></tr><tr><td colspan="2">Side bearer springs are condemned on the basis of height. It is recommended springs having variation up to 2 mm in free height are only assembled in the same bolster. Mixing of new and old spring should be avoided.</td><td>123</td><td>119</td></tr></table> <table><tr><th>H</th><th>Wedge Block</th><th>Wear limit</th></tr><tr><td>i).</td><td>Wear in Vertical Surface from C/L of Spigot.</td><td>7</td></tr><tr><td>ii)</td><td>Slope Surface by Gauge</td><td>3</td></tr></table>	B	PEDESTAL CROWN	Nominal	(Wear Limit)	i)	Key Seat to Pedestal	318	5	ii)	Pedestal Crown Side	152	4+4	iii)	Pedestal Side	105	2+2	iv)	Distance between Outer & Inner Pedestal Jaw	278	4+4	v)	Crown side	152	4+4	C	ADAPTER	Nominal	(Wear Limit)	i)	Crown Lug	156	4+4	ii)	Side Lug	130	3+3	iii)	Adapter Side	268	3+3	D.	BOGIE BOLSTER. Bolster pocket has 8 mm thick Si-Mg steel liner in slope pocket, may be permitted in service up to 3mm thick. Replace worn liner. No paint or grease should be applied on liner.	Nominal	(Wear Limit)	i)	Bolster Land Surface	443	3+3	ii)	Thickness of Slope Surface Liner	8	3+3	iii)	Rotation Stop Lug	518	3+3	iv)	Bolster Column Jib Outer	241	5+5	v)	Bolster Column Jib Inner	136	5+5	E.	Center Pivot	(Wear Limit)	i).	Wear Limit Vertical side	2	ii)	Seat.	2	F	Elastomeric Pad	i).	The top or the bottom plates should not show any cracks. If any sign of crushing of rubber is noticed reject the pad.	ii)	Permanent set in free height is allowed to 42 mm instead of 46 mm normal size. Use up-to 44 mm in ROH	G	Side bearer	Nom	Cond	Side bearer springs are condemned on the basis of height. It is recommended springs having variation up to 2 mm in free height are only assembled in the same bolster. Mixing of new and old spring should be avoided.		123	119	H	Wedge Block	Wear limit	i).	Wear in Vertical Surface from C/L of Spigot.	7	ii)	Slope Surface by Gauge	3		
B	PEDESTAL CROWN	Nominal	(Wear Limit)																																																																																																
i)	Key Seat to Pedestal	318	5																																																																																																
ii)	Pedestal Crown Side	152	4+4																																																																																																
iii)	Pedestal Side	105	2+2																																																																																																
iv)	Distance between Outer & Inner Pedestal Jaw	278	4+4																																																																																																
v)	Crown side	152	4+4																																																																																																
C	ADAPTER	Nominal	(Wear Limit)																																																																																																
i)	Crown Lug	156	4+4																																																																																																
ii)	Side Lug	130	3+3																																																																																																
iii)	Adapter Side	268	3+3																																																																																																
D.	BOGIE BOLSTER. Bolster pocket has 8 mm thick Si-Mg steel liner in slope pocket, may be permitted in service up to 3mm thick. Replace worn liner. No paint or grease should be applied on liner.	Nominal	(Wear Limit)																																																																																																
i)	Bolster Land Surface	443	3+3																																																																																																
ii)	Thickness of Slope Surface Liner	8	3+3																																																																																																
iii)	Rotation Stop Lug	518	3+3																																																																																																
iv)	Bolster Column Jib Outer	241	5+5																																																																																																
v)	Bolster Column Jib Inner	136	5+5																																																																																																
E.	Center Pivot	(Wear Limit)																																																																																																	
i).	Wear Limit Vertical side	2																																																																																																	
ii)	Seat.	2																																																																																																	
F	Elastomeric Pad																																																																																																		
i).	The top or the bottom plates should not show any cracks. If any sign of crushing of rubber is noticed reject the pad.																																																																																																		
ii)	Permanent set in free height is allowed to 42 mm instead of 46 mm normal size. Use up-to 44 mm in ROH																																																																																																		
G	Side bearer	Nom	Cond																																																																																																
Side bearer springs are condemned on the basis of height. It is recommended springs having variation up to 2 mm in free height are only assembled in the same bolster. Mixing of new and old spring should be avoided.		123	119																																																																																																
H	Wedge Block	Wear limit																																																																																																	
i).	Wear in Vertical Surface from C/L of Spigot.	7																																																																																																	
ii)	Slope Surface by Gauge	3																																																																																																	

SN	Requirements	Clause	Observations																
	<table><tr><th>I</th><th>Bogie Dimension</th><th>Nominal</th><th>Tolerance</th></tr><tr><td>i)</td><td>Cross Trammel</td><td>3018</td><td>±4.5</td></tr><tr><td>ii)</td><td>Journal Center Distance</td><td>2060</td><td>±5</td></tr></table>	I	Bogie Dimension	Nominal	Tolerance	i)	Cross Trammel	3018	±4.5	ii)	Journal Center Distance	2060	±5						
I	Bogie Dimension	Nominal	Tolerance																
i)	Cross Trammel	3018	±4.5																
ii)	Journal Center Distance	2060	±5																
1.3.4	Brake gear attention is given as under: <table><tr><th></th><th>Brake Gear Repairs</th></tr><tr><td>i)</td><td>SAB (set 'A' dimension to 72 +2, -0 mm)</td></tr><tr><td>ii)</td><td>Control Rod – Replace on condition basis</td></tr><tr><td>iii)</td><td>LSD (maintain gap 8 to 10 cm)</td></tr><tr><td>iv)</td><td>Hand Brake Wheel – operate and attend</td></tr><tr><td>v)</td><td>Hand Brake Connecting Rod – check and replace on need basis</td></tr><tr><td>vi)</td><td>Ext. Roller Bracket – check and replace on need basis</td></tr><tr><td>vii)</td><td>Pins & Cotters. – check and replace on need basis</td></tr></table>		Brake Gear Repairs	i)	SAB (set 'A' dimension to 72 +2, -0 mm)	ii)	Control Rod – Replace on condition basis	iii)	LSD (maintain gap 8 to 10 cm)	iv)	Hand Brake Wheel – operate and attend	v)	Hand Brake Connecting Rod – check and replace on need basis	vi)	Ext. Roller Bracket – check and replace on need basis	vii)	Pins & Cotters. – check and replace on need basis		
	Brake Gear Repairs																		
i)	SAB (set 'A' dimension to 72 +2, -0 mm)																		
ii)	Control Rod – Replace on condition basis																		
iii)	LSD (maintain gap 8 to 10 cm)																		
iv)	Hand Brake Wheel – operate and attend																		
v)	Hand Brake Connecting Rod – check and replace on need basis																		
vi)	Ext. Roller Bracket – check and replace on need basis																		
vii)	Pins & Cotters. – check and replace on need basis																		
1.3.5	Wheel sets and CTRB's are attended as per para 1.2.2 (g)																		
1.3.6	CBC and DG are attended as per para 1.2.2 (f)																		
1.3.7	Air brake system is attended as per para 1.2.2. (d)																		
1.3.8	Welding preparation: a) The surface to be welded should be cleaned properly to remove dust , oil, cracks, scales preferably by grinding with portable grinder b) All welding shall be carried out in down hand position with the help of manipulator whenever required. c) Low hydrogen electrodes approved by RDSO shall be used. d) For welding of liners start from one corner and then weld opposite corner and so on till four corners are completed in the same pass. e) Electrodes shall be pre heated in a drying oven at temperature of 150-200°C at least one hour before use. f) No post weld heat treatment is necessary. g) Reconditioned area shall be examined visually as well as by DTP to ensure freedom from any defect / cracks.																		
1.3.9	ASSEMBLY: a) Lower the under frame on bogie provide pivot pin shackle, pin with split pin and connect brake gear pin. Dimension-60 between CP top to side bearer top liner should be maintained. b) Check distance of side frame top liner to side bearer keep distance within tolerance (131 +0, -0.5) by altering. c) Ensure all APD measure are incorporated after assembly. d) Paint bogie, stencil station code and date of ROH. e) Draw bar height and CBC height should not be more than 845 mm and 1105 mm respectively. No packing should be given over axle box adapter. f) Check proper fitment operation of twist lock for proper locking.																		

SN	Requirements	Clause	Observations																						
1.3.10	After assembly and completion of finishing work, carry out air brake testing with SWTR <table><tr><th>Check point</th><th>Specified</th></tr><tr><td>i). BP Pressure</td><td>5 ± 0.1kg/cm²</td></tr><tr><td>ii). AUX. Reservoir Pressure</td><td>5 ± 0.1kg/cm²</td></tr><tr><td>iii) Leakage from system after charging</td><td>0.1kg/cm² in one minute</td></tr><tr><td>iv) Full service and emergency application cylinder filling time</td><td>18 to 30 seconds</td></tr><tr><td>v). Maximum brake cylinder pressure</td><td>Empty 2.2 ± 0.25 kg/cm² Loaded 3.8 ± 0.1 kg/cm²</td></tr><tr><td>vi). Reduction in BP Pressure required for full service application</td><td>1.3 to 1.6 kg/cm²</td></tr><tr><td>vii). Release after full service</td><td>45 to 60 seconds</td></tr><tr><td>viii). Sensitivity of brakes. Isolate brake from mail line. Check response of brakes when BP Pressure is reduced to 0.6 kg/cm² in 6 seconds.</td><td>Brake should apply within 6 seconds.</td></tr><tr><td>ix). Insensitivity of brakes. Isolate brake pipe from mail line. Check response of brakes when BP Pressure reduced to 0.3 kg/cm² in 60 seconds</td><td>Brake should not apply</td></tr><tr><td>x). Piston stroke</td><td>Empty 85 ± 10 mm Loaded 120 ± 10</td></tr></table>	Check point	Specified	i). BP Pressure	5 ± 0.1kg/cm ²	ii). AUX. Reservoir Pressure	5 ± 0.1kg/cm ²	iii) Leakage from system after charging	0.1kg/cm ² in one minute	iv) Full service and emergency application cylinder filling time	18 to 30 seconds	v). Maximum brake cylinder pressure	Empty 2.2 ± 0.25 kg/cm ² Loaded 3.8 ± 0.1 kg/cm ²	vi). Reduction in BP Pressure required for full service application	1.3 to 1.6 kg/cm ²	vii). Release after full service	45 to 60 seconds	viii). Sensitivity of brakes. Isolate brake from mail line. Check response of brakes when BP Pressure is reduced to 0.6 kg/cm ² in 6 seconds.	Brake should apply within 6 seconds.	ix). Insensitivity of brakes. Isolate brake pipe from mail line. Check response of brakes when BP Pressure reduced to 0.3 kg/cm ² in 60 seconds	Brake should not apply	x). Piston stroke	Empty 85 ± 10 mm Loaded 120 ± 10		
Check point	Specified																								
i). BP Pressure	5 ± 0.1kg/cm ²																								
ii). AUX. Reservoir Pressure	5 ± 0.1kg/cm ²																								
iii) Leakage from system after charging	0.1kg/cm ² in one minute																								
iv) Full service and emergency application cylinder filling time	18 to 30 seconds																								
v). Maximum brake cylinder pressure	Empty 2.2 ± 0.25 kg/cm ² Loaded 3.8 ± 0.1 kg/cm ²																								
vi). Reduction in BP Pressure required for full service application	1.3 to 1.6 kg/cm ²																								
vii). Release after full service	45 to 60 seconds																								
viii). Sensitivity of brakes. Isolate brake from mail line. Check response of brakes when BP Pressure is reduced to 0.6 kg/cm ² in 6 seconds.	Brake should apply within 6 seconds.																								
ix). Insensitivity of brakes. Isolate brake pipe from mail line. Check response of brakes when BP Pressure reduced to 0.3 kg/cm ² in 60 seconds	Brake should not apply																								
x). Piston stroke	Empty 85 ± 10 mm Loaded 120 ± 10																								

INFRASTRUCTURAL FACILITIES IN ROH DEPOTS

(Ref: Appendix –II of CAMTECH Wagon manual)

A) Depot Name:

B) Category of depot:

Category	Targeted capacity for ROH (Average/month)	Status of Depot (Av ROH attention per month in VU during last one year)
Super Depots	Above 500	
Mega Depots	250 to 500	
Major Depots	125 to 250	
Minor Depots	Upto 125	

C) Covered area under cranes: The recommended crange area for depots undertaking 250 to 500 ROHs per month is given below:

1. For ROH depots WITH sick-line attention:

Requirements			Observations
250 ROHs/month + sick line repairs with double shift.		5000 sq. m including wheel lathe shed	
300	ROHs/month + sick line repairs with double shift.	5600 sq. m	
350		6200 sq. m.	
400		6800 sq. m	
450		7400 sq. m	
500		8000 sq. m	

2. For ROH depots WITHOUT sick-line attention:

Requirements	Observations
Depots undertaking only ROH workload and no out-of-course repairs can manage the outturn of 250 ROHs/month in about 3000 sq. m. crange area with working in double shifts.	

D) M&P requirement:

Recommended list of M&P for a unit depot i.e. handling 250 ROH per month as per CAMTECH wagon manual. For a depot handling more than 250 ROH per month, the requirement has to be scaled up depending on depot layout and facilities created. Requirement of M&P for each section is given below:

1. Body section

SN	M&P	Quantity	Observations
1	EOT cranes 15t/5t	2 Nos	
2	Welding machines	4 Nos	
3	Portable hydraulic rivetter	1 No	
4	Trestles	12 sets	
5	PC terminal (common with bogie section if layout permits)	No. will depend on layout.	
6	Portable Winches	As per layout	
7	Grit Blasting Machine	1 No.	

2. Bogie section

SN	M&P	Quantity	Observations
1	EOT crane 10t / 3t	2 Nos	
2	Welding machines	6 Nos	
3	Portable hydraulic rivetter	1 Nos	
4	Stores bin	20 Nos	
5	Portable electric grinders	3 Nos.	
6	Jib crane 2.5t	10 Nos.	
7	Fixture for rivetting spring plank	2 Nos.	
8	Work station for bogie repair	10 Nos.	
9	Turn table for wheel sets	As per layout	
10	Road crane 10t capacity	1 No	
11	Bogie manipulators	3 Nos.	
12	Roller bearing diagnostic equipment	2 Nos.	
13	Magnetic flaw detector	2 Nos.	
14	CO2 welding machine	2 Nos.	

3. Air brake equipment

SN	M&P	Quantity	Observations
1	Elec. Stationery screw air compressors complete with air receiver 15 m 3/min (This is for depot only. Additional compressors will be required for yard).	1 No.	
2	Portable diesel compressor	1 No.	
3	Single Wagon test rig	2 Nos.	
4	DV test stand	1 No	
5	Hydraulic pipe bending machine	1 No.	
6	Air conditioners with voltage stabilizer	2 Nos.	
7	Torque wrench with various sizes of sockets less than 1 inch	4 set.	
8	Ultrasonic cleaning table for DV components	1 Nos.	

4. Wheel Reconditioning Equipment

SN	M&P	Quantity	Observations
1	Surface wheel lathe with Servo controlled voltage stabilizer	1 No.	
2	Pneumatic torque wrench with sockets 1" to 2"	4 sets	
3	Pressure grease drum with guns 15t capacity.	2 Nos.	
4	Ultrasonic flaw detector	2 Nos	

5	Bearing marking gadget	1 set	
6	Mono-rail for handling swarf (detail plan will depend on layout)	1 set	
7	7 Bins for storing bearings	As per requirement	
8	Fork lifter 2.5t	1 No.	
9	P.C. Terminal	1 No.	
10	Wheel diameter measuring gauge	2 Nos.	

5. Machine shop & material reclamation

SN	M&P	Quantity	Observations
1	Heavy duty shaping machine	1 No	
2	Jib crane 2.5t	1 No	
3	Centre lathe 12 ½" cap.	1 No.	
4	Radial drilling machine	1 No.	
5	Centre lathe 6" cap	1 No.	
6	Bench drilling machine	1 No.	
7	Heavy duty pedestal grinder double ended	1 No.	
8	Shearing machine 6 mm	1 No.	

6. Stores

SN	M&P	Quantity	Observations
1	Truck 10t capacity	1 No.	
2	Tractor with hydraulic lifting machine and 3 trolleys (trailers)	1 No.	
3	Fork lift 2t	2 No.	
4	Platform truck 2t	2 Nos.	
5	Battery charger	2 No.	
6	Weighing machine 500 kg	1 No.	
7	Light store vehicle	1 No.	
8	Hand trolleys with rubber wheels	10 Nos.	
9	Storage racks	As per layout	
10	Computer terminal with printer	1 No.	
11	Furniture	As per layout	

7. Black smithy

SN	M&P	Qunatity	Observations
1	Black smithy hearth	1 No.	
2	Anvil	1 No.	
3	Hand tools	5 sets	
4	Motorised blower	1 No.	

8. Training equipment

SN	M&P	Quantity	Observations
1	Cut models of DV, SAB, Angle cock.	1 No. of each design	
2	Television (large screen) projector type	1 No.	
3	VCR	1 No.	
4	Furniture	As per layout	
5	Hostel along with kitchen equipment		
6	Overhead projector	1 No.	
7	LCD projector	1 No	

9. Canteen & staff amenities

SN	M&P	Quantity	Observations
1	Water cooler	1 3 No.	
2	Stainless steel tables & chairs	As per reqt.	
3	Kitchen equipment, cooking gas, utensils	As per reqt.	
4	Fitter's lockers	As per reqt.	

10. Office Equipment

SN	M&P	Quantity	Observations
1	PC terminal	2 No.	
2	Printer	1 No.	
3	Furniture	As per layout	
4	Air conditioner for DME's chamber & computer room.	3 No.	
5	Intercom 10 lines	1 No.	
6	Fax with P&T line	1 No.	
7	First aid equipment	4 set	
8	Photocopier	1 No.	
9	V.H.F. sets.	10 sets	

11. Miscellaneous

SN	M&P	Quantity	Observations
1	Gas cutting equipment	4 Nos.	
2	Hydraulic jacks	As per requt	
3	SAB test bench	1 No.	
4	CBC and draft gear replacement equipment	1 No.	
5	DG set 350/500 KVA	1 No.	
6	Lumpsum for electric and pneumatic tools		
7	Lumpsum for fitter's hand tools.		
8	Lumpsum for gauges & instruments		
9	Wheel flat detector system *		
10	Wheel flange welding machine *		

(*) To be installed in depots/yards as per instructions of Railway Board.

12. Where power failures are rampant, a Diesel Generating Set to run the wheel lathe and cranes may also be planned.

Quality Assurance Directorate (Mechanical)
RDSO Lucknow

**Quality audit check-sheet for Yard attention (Intensive examination) and
sick-line attention of air brake freight stock**

(Based on CAMTECH Maintenance Manual for Wagon and CAMTECH report on
Standardisation of Infrastructure Facilities for Freight stock maintenance)

SN	Requirements	Clause	Observations
1	Railway is following comprehensive instructions regarding the pattern of freight train examination and issue of brake power certificate issued by Railway Board.	RB letter no. 94/M(N)/951/57/Vol II/Pt dtd 25.10.04	
1.1	Zonal Railways should issue a detailed joint inspection for C&W examination for freight train in terms of RB guidelines by Operating and Mechanical department every year in line with changes in traffic pattern	RB letter no. 78 / M(W)/814/8 Vol.II dtd 01.10.80 and 91/ M(N)/951/31 dtd 24.03.95	
2	Intensive examination of different stock: a) All freight trains should be subjected to intensive examination in empty condition at originating station. b) In exceptional cases, the back loaded freight train can be examined as per instruction mentioned in para 314 item (v) of CAMTECH Wagon manual. c) All freight trains shall be re-examined if stabled for more than 24 hrs by JE (C&W) in yard and by Guard and Driver in non C&W station up to next C&W point in the direction of movement for examination. d) Air brake stock should run on end to end pattern / premium / CC pattern. The intensive BPC shall remain valid provided: ➤ The destination is clearly mentioned on the BPC of loaded train. ➤ The composition of rake is not changed by 4 or more wagons (5 or more wagon for BLC rake.)	Para 303 of CAMTECH Wagon manual	
3	Premium examination of air brake freight train : a) Premium rakes will be intensively examined in empty condition and certified by examination point nominated by CME & COM. Such premium examination point should be "A" category" or should be upgraded to 'A' category. b) If any of the condition i.e. examination in empty condition or examination at nominated point is not satisfied then rake will operate normal end to end rake. c) BPC issued for such premium rake will be valid for 12 days from the date of issue (Grace period of 3 days for unloading and to avoid examination in loaded condition) d) After each loading / unloading, the rake will be examined by Guard and driver before commencement of journey and recorded under the relevant column of BPC. In case of mechanized loading / Unloading, examination by TXR is desirable. e) Premium rake should be turned out with 95% Brake Power.	RB letter no. 2005/M(N)/951/13 dtd 07.04.06	

SN	Requirements	Clause	Observations
	f) Movement of premium rake is monitored through FIOS.		
3.1	All points nominated for premium examination of trains shall issue premium intensive BPC and such rakes will be formed by all type of air brake freight rolling stock except BRN, BTPGLN, BLC, BFKN & departmental stock. BLC rake will be examined on CC basis.	RB letter no. 2005/M(N)/ 951/13 / Pt dtd 31.01.07	
4.0	Close Circuit pattern of examination:		
4.1	BOXNHL and BCNHL rakes are examined on CC pattern with the following condition: ➤ CC rake of BOXNHL and BCNHL will have validity of 7500 Km/ 35 days. ➤ Rakes are examined at nominated point. During CC examination, proper attention to the safety fittings shall be given and their maintenance records shall be maintained separately. ➤ CC examination will be done only at examination lines in the yard having proper facility of material handling, concrete pathway, welding, lighting etc. The CC examination will be done lines on which there is no OHE. ➤ CC rake is monitored through FOIS. In case CC rake is not mad available for examination at the base depot and offered for examination at some other depot, it will become non CC rake and it will follow the normal intensive examination pattern. ➤ After each loading/unloading, the rake will be examined by Guard and Driver and observations recorded on BPC. ➤ Detail record should be kept for the performance of the rakes and their condition should be recorded during every examination.	RB letter no. 2008/M(N)/ 951/31 CC Rakes dtd 11.03.10	
4.2	CC examination of other air brake freight stock: ➤ The air brake CC rake shall be offered for periodical maintenance examination (PME) in empty condition at the CC depot that formed CC rake originally. ➤ Rake integrity as listed in BPC should be maintained. However, upto 4 wagons (for BLC rake – 5 wagons) may be replaced by good examined wagons in the entire run between two PMEs. ➤ The KMs have been logged in correctly and continuously. ➤ The rake is not stabled for more than 24 hrs. ➤ The rake is running in predefined circuit as mentioned in BPC. ➤ After each loading/unloading, the rake will be examined by Guard and Driver and observations recorded on BPC. ➤ All CC freight trains will be given intensive examination during day light hours. ➤ BPC of CC rake is valid for 7500 km / 35 days for 'A' category points (as notified) and 6000 km/30 days for other examination point as per JPO issued by Zonal Railways. ➤ POH / ROH wagons from CC rakes will be marked and detached at base depots only. ➤ CC rake should be preferably formed by off ROH / POH wagons	RB letter no. 94/M(N)/ 951/57/Vol II/Pt dtd 25.10.04 and para 303 of CAMTECH Wagon manual.	
4.2.1	BOXN / BCN CC rakes are to be formed at nominated examination points.	RB letter no. 2008/M(N)/ 951/13/CC	

SN	Requirements	Clause	Observations
		rakes dtd 28.10.09	
4.2.2	BTPN CC rakes are to be formed at nominated examination points	RB letter no. 2008/M(N)/ 951/13/CC rakes dtd 05.02.10 & 13.07.10	
5.0	Attention to be given during intensive examination (CC/ premium / end to end):		
5.1	Adequate gang strength is deployed for intensive examination: Air brake (end to end): 56 man-hour Air brake (CC): 100 man-hour Premium rake: 66 man-hour	Para 313 of CAMTECH Wagon manual CAMTECH letter no. IR/CAMTECH. M. GWL.Premium Fr Dtd 29.05.09	
5.2	Rolling-in examination including axle box feeling: ➤ All terminating trains should be given rolling-in examination while entering a Station / Yard with a train examination depot. ➤ The following inspection should be carried during the rolling –in examination: ✓ In motion inspection and observation of under gear of wagons for any loose or dangling components and flat places on tyres / wheels ✓ After train has come to halt, all boxes should be felt within 20 mts. ✓ Examination of any abnormal behaviour of any of the vehicles or any other observation which may relate to unsafe working condition. ✓ The rolling –in examination must be conducted to detect any skidded wheel. ✓ In-coming BPC should be collected by Yard C&W staff.	Para 305A of CAMTECH Wagon manual.	
5.2	Intensive examination and repairs: Before undertaking intensive examination, the rake should be protected at both the ends. The following examination and repair activities should be carried out: ➤ Inspection and repairs of running gear fittings. ➤ Inspection and repairs of brake gear and spring gears. ➤ Inspection and repairs of draw and buffing gear. ➤ Checking and making good the deficiency of safety fittings, safety bracket, safety loop etc. ➤ Replacement of brake block on condition basis. ➤ Correct fitment of washers, bulb cotters and all brake gear pins to be ensured. ➤ Correct functioning and positioning of empty load device. ➤ Wagon with warm axle box to be detached and may be taken in sick line for further action. ➤ Checking and proper securing doors. ➤ Observe CBC height, wear plate , Knuckle etc by visual examination. In case of doubt, buffer height should be measured.	Para 305 B of CAMTECH Wagon manual	

SN	Requirements	Clause	Observations									
	➤ Check brake cylinders, DV, AR, CR and other pipe joints to ensure that these are in proper working order. Isolating cock and angle cock to be checked for proper position. Brake cylinder should be released and checked for piston stroke for empty and loaded condition. ➤ After brakes are released, the wheel profile should be examined visually. If any defect is noticed, it should be checked with tyre defect gauge and wagon to be marked sick for wheel changing. ➤ The bogies complete side frames and bolster to be visually examined for cracks and missing parts, bolster springs, snubber, spigot, centre pivot fastening etc to be checked for defects and attended. ➤ Examine brake rigging components with special attention to brake beam deformation and wear on integral brake shoe bracket. Check pull rod, push rod with pins, washers, split pins and cotters. ➤ Hand brake must be checked for smooth and effective operation. ➤ Visual examination of under-frame members, body, door mechanism, CBC wear or deficiency of parts to be marked and their operation to be checked. ➤ Brake power should be ensured. (CC – 100%, Premium end to end – 95 % and end to end examination – 90 %) ➤ Where a reject-able defect cannot be attended in Yard, the wagon shall be damaged for attention in sick line.											
5.3	Issue of Brake Power Certificate: ➤ All freight trains after intensive examination will be given a Brake Power certificate. ➤ BPC certificate should be printed in different colour on good quality paper: ✓ CC rake – Yellow colour. ✓ Premium rake – Green colour ✓ Air brake end to end – Green colour. ➤ BPC should be issued in standard format. ➤ Originating brake power for air brake rake in CC shall be 100% ➤ Originating brake power for premium air brake rake shall be minimum 95% ➤ The minimum brake power for air brake freight train on end to end pattern shall be 90%. ➤ No fresh BPC shall be issued during re-validation. ➤ No safe to run BPC issued from nodal point. ➤ Brake pipe pressure in air brake train with locomotive should be as follows: <table><tr><td>Nos of wagons</td><td>On locomotive</td><td>On last wagon</td></tr><tr><td>Upto 56</td><td>5.0 Kg/Cm²</td><td>4.8 Kg/Cm²</td></tr><tr><td>Beyond 56</td><td>5.0 Kg/Cm²</td><td>4.7 Kg/Cm²</td></tr></table>	Nos of wagons	On locomotive	On last wagon	Upto 56	5.0 Kg/Cm ²	4.8 Kg/Cm ²	Beyond 56	5.0 Kg/Cm ²	4.7 Kg/Cm ²	Para 305 C of CAMTECH wagon manual.	
Nos of wagons	On locomotive	On last wagon										
Upto 56	5.0 Kg/Cm ²	4.8 Kg/Cm ²										
Beyond 56	5.0 Kg/Cm ²	4.7 Kg/Cm ²										
5.4	Important parameters to be ensured during intensive examination:	Para 307 of CAMTECH wagon manual										

SN	Requirements	Clause	Observations																		
5.4.1	Brake gear limit and clearances: <table><tr><td>Description</td><td>Limit</td></tr><tr><td>Brake block cond limit</td><td>10 mm</td></tr><tr><td>Yard leaving thickness of brake block except BOY</td><td>20 mm</td></tr><tr><td>A dimension of air brake stock fitted with CASNUB bogie except BOBRN</td><td>70 mm + 2 -0</td></tr><tr><td>A dimension of BOBRN wagons</td><td>27 mm + 2 -0</td></tr></table>	Description	Limit	Brake block cond limit	10 mm	Yard leaving thickness of brake block except BOY	20 mm	A dimension of air brake stock fitted with CASNUB bogie except BOBRN	70 mm + 2 -0	A dimension of BOBRN wagons	27 mm + 2 -0										
Description	Limit																				
Brake block cond limit	10 mm																				
Yard leaving thickness of brake block except BOY	20 mm																				
A dimension of air brake stock fitted with CASNUB bogie except BOBRN	70 mm + 2 -0																				
A dimension of BOBRN wagons	27 mm + 2 -0																				
5.4.2	Piston stroke: <table><tr><td rowspan="2">Types of wagon</td><td colspan="2">Piston stroke</td></tr><tr><td>Empty</td><td>Loaded</td></tr><tr><td>BOXN, BCN, BRN, BTPGLN</td><td>85 mm +/- 10</td><td>130 mm +/- 10</td></tr><tr><td>BTPN</td><td>87 mm +/- 10</td><td>117 mm +/- 10</td></tr><tr><td>BOY</td><td>90 mm +/- 10</td><td>135 mm +/- 10</td></tr><tr><td>BOBRN/ BOBR</td><td>100 mm +/- 10</td><td>110 mm +/- 10</td></tr></table>	Types of wagon	Piston stroke		Empty	Loaded	BOXN, BCN, BRN, BTPGLN	85 mm +/- 10	130 mm +/- 10	BTPN	87 mm +/- 10	117 mm +/- 10	BOY	90 mm +/- 10	135 mm +/- 10	BOBRN/ BOBR	100 mm +/- 10	110 mm +/- 10			
Types of wagon	Piston stroke																				
	Empty	Loaded																			
BOXN, BCN, BRN, BTPGLN	85 mm +/- 10	130 mm +/- 10																			
BTPN	87 mm +/- 10	117 mm +/- 10																			
BOY	90 mm +/- 10	135 mm +/- 10																			
BOBRN/ BOBR	100 mm +/- 10	110 mm +/- 10																			
5.4.3	Buffer Height from Rail Level: ➤ Maximum 1105 mm (Empty) ➤ Minimum 1030 mm (Loaded)																				
6.0	Important parameters to be ensured during Sick line / Depot attention:	Para 308 of CAMTECH wagon manual																			
6.1	Nominal clearances of CASNUB bogie: <table><tr><td>Description</td><td>22NLB</td><td>22 HS</td></tr><tr><td>Lateral clearance between side frame and bolster</td><td>18 mm</td><td>25 mm</td></tr><tr><td>Lateral clearance between side frame and axle box/adopter</td><td>16 mm</td><td>16 mm</td></tr><tr><td>Longitudinal clearance between side frame and axle box / adopter</td><td>9 mm</td><td>9 mm</td></tr><tr><td>Longitudinal clearance between side frame and bolster</td><td>6 mm</td><td>6 mm</td></tr><tr><td>Clearance between anti rotation lug and bolster</td><td>4 mm</td><td>4 mm</td></tr></table>	Description	22NLB	22 HS	Lateral clearance between side frame and bolster	18 mm	25 mm	Lateral clearance between side frame and axle box/adopter	16 mm	16 mm	Longitudinal clearance between side frame and axle box / adopter	9 mm	9 mm	Longitudinal clearance between side frame and bolster	6 mm	6 mm	Clearance between anti rotation lug and bolster	4 mm	4 mm	Para 308 C of CAMTECH wagon manual	
Description	22NLB	22 HS																			
Lateral clearance between side frame and bolster	18 mm	25 mm																			
Lateral clearance between side frame and axle box/adopter	16 mm	16 mm																			
Longitudinal clearance between side frame and axle box / adopter	9 mm	9 mm																			
Longitudinal clearance between side frame and bolster	6 mm	6 mm																			
Clearance between anti rotation lug and bolster	4 mm	4 mm																			

SN	Requirements	Clause	Observations																														
6.2	Wear limits <table><tr><td>Adapter thrust shoulder</td><td>0.7 mm</td></tr><tr><td>Adapter crown lugs</td><td>4.0 mm</td></tr><tr><td>Adapter crown seat</td><td>3.5 mm</td></tr><tr><td>Adapter side lugs</td><td>3.0 mm</td></tr><tr><td>Adapter sides</td><td>3.0 mm</td></tr><tr><td>Side frame column friction plate</td><td>4.0 mm</td></tr><tr><td>Side frame column sides</td><td>5.0 mm</td></tr><tr><td>Side frame anti rotation lug</td><td>3.0 mm</td></tr><tr><td>Pedestal crown roof</td><td>5.0 mm</td></tr><tr><td>Pedestal crown sides</td><td>4.0 mm</td></tr><tr><td>Pedestal sides</td><td>2.0 mm</td></tr><tr><td>Pedestal jaw</td><td>4.0 mm</td></tr><tr><td>Bolster liner wear limit</td><td>5.0 mm</td></tr><tr><td>Bolster land surface</td><td>3.0 mm</td></tr><tr><td>Bolster column sides – Inner / outer</td><td>5.0 mm</td></tr></table>	Adapter thrust shoulder	0.7 mm	Adapter crown lugs	4.0 mm	Adapter crown seat	3.5 mm	Adapter side lugs	3.0 mm	Adapter sides	3.0 mm	Side frame column friction plate	4.0 mm	Side frame column sides	5.0 mm	Side frame anti rotation lug	3.0 mm	Pedestal crown roof	5.0 mm	Pedestal crown sides	4.0 mm	Pedestal sides	2.0 mm	Pedestal jaw	4.0 mm	Bolster liner wear limit	5.0 mm	Bolster land surface	3.0 mm	Bolster column sides – Inner / outer	5.0 mm	Para 308 D of CAMTECH wagon manual	
Adapter thrust shoulder	0.7 mm																																
Adapter crown lugs	4.0 mm																																
Adapter crown seat	3.5 mm																																
Adapter side lugs	3.0 mm																																
Adapter sides	3.0 mm																																
Side frame column friction plate	4.0 mm																																
Side frame column sides	5.0 mm																																
Side frame anti rotation lug	3.0 mm																																
Pedestal crown roof	5.0 mm																																
Pedestal crown sides	4.0 mm																																
Pedestal sides	2.0 mm																																
Pedestal jaw	4.0 mm																																
Bolster liner wear limit	5.0 mm																																
Bolster land surface	3.0 mm																																
Bolster column sides – Inner / outer	5.0 mm																																
6.3	Free height and condemning limit for CASNUB bogie springs (Outer, Inner and Snubber) are maintained as recommended in para 308 E	Para 308 E of CAMTECH wagon manual																															
6.4	Wear limit for Friction wedge is maintained as recommended	Para 308 F of CAMTECH wagon manual																															
6.5	Centre pivot dimension is checked as recommended	Para 308 G of CAMTECH wagon manual																															
6.6	Wheels and axles are checked for condemning diameter	Para 308 I of CAMTECH wagon manual																															
7.0	Air brake testing facility: i) Compressor of adequate capacity to carry out air brake examinations of rake during intensive examination as well as air brake testing of wagon in sick line . Pneumatic line is provided with moisture trap or air-dryer to ensure availability of moisture free air for air brake testing. ii) Availability of adequate RTR for air brake testing during intensive examination and SWTR for air brake testing in sick line.	Ch 3 of CAMTECH Manual for Wagon																															
8.0	Infrastructure facility for Yard maintenance i) Adequate centre to centre distance between tracks (at least 7.5 mts) for conducting intensive examination. ii) Concrete pathway from one end to other end. Material handling facility to ensure movement of material from one end to other end. iii) Proper illumination for conducting intensive examination. iv) Welding facility (Rectifier type) for welding during intensive examination. v) Adequate outlets for tapping air pressure for testing of stock. vi) Duty room for maintenance supervisor, compressor room, store room, tool room, welding machine room for intensive examination. vii) VHF sets for close monitoring. Infrastructure facilities for Yard attention to be verified as per Checklist (Annexure-I) of CAMTECH guidelines for Standardization of Infrastructure facilities for freight stock maintenance	Ch 3 of CAMTECH Manual for Wagon CAMTECH report on Standardisation of Infrastructure facilities for freight stock maintenance CAMTECH / 2003/M/NF-STD Aug 2003																															

SN	Requirements	Clause	Observations				
9	Sick line facilities: Infrastructure facilities should be available as per Table II of CAMTECH guidelines for Standardization of Infrastructure facilities for freight stock maintenance	CAMTECH report on Standardisation of Infrastructure facilities for freight stock maintenance CAMTECH / 2003/M/NF-STD Aug 2003					
10	Analysis of sick marking and yard attention is being carried out. Computer facility for such analysis is available.	-					
11	General housekeeping: <table border="1"><tr><td>Level of Housekeeping</td></tr><tr><td>Overall ambience and working condition</td></tr><tr><td>Storage of Wheel sets</td></tr><tr><td>Storage of Air brake cylinder, BMBS, AR and other stores.</td></tr></table>	Level of Housekeeping	Overall ambience and working condition	Storage of Wheel sets	Storage of Air brake cylinder, BMBS, AR and other stores.		
Level of Housekeeping							
Overall ambience and working condition							
Storage of Wheel sets							
Storage of Air brake cylinder, BMBS, AR and other stores.							
12.0	Miscellaneous items for attention as advised by RB						
12.1	Corrective and preventive measure to detect hot axle box cases (potential case of train accident due hot axle breakage) ➤ Alertness of TXR staff on platforms, station staff, gateman, signal maintainers etc for detecting hot axle box case and taking necessary corrective action. ➤ Safety poster depicting hot box detection and immediate follow up should be affixed at all stations, gate huts, cabins and maintenance depot. ➤ Ensure practice of box feeling immediately on arrival of train in examination yard. Measure axle box temperature with non contact measuring temperature instrument. ➤ Care should be taken at wagon depot while handling wheel sets to avoid any damage to grease seals. ➤ All cases of hot BOX must be investigated and a detailed report must be submitted to RDSO.	RB letter no. 2005/M(N)/951/49 dtd 05.10.11					
12.2	Attention for non –functional hand brake: ➤ RB has advised Railway to take immediate remedial measure to make good non-functional hand brake. ➤ Action taken by Railway for detecting non functional hand brake and making good deficiency in Workshop, ROH depot, Sick-lines and Yard.	RB letter no. 2012/M(Safety)/10/ Misc Inspection dtd 05.10.11					
12.3	Action taken by Railway on the basis of CRS/NE's circle final report: ➤ Para 9.1.1 of CRS report: Railway should carry out a drive to check the fitment of hand brake wheel and its spindle on the wagons. Riveting of spindle with sleeves should be done wherever deficient in accordance with RDSO drg no. WD – 09056 – S - 07. ➤ Para 9.1.2 of CRS report: In addition to riveting, welding of spindle with sleeve should also be considered as this item is said to be prone to theft.	RB letter no. 2012/M(Safety)/7/1 NCR-II dtd 24.05.12					

SN	Requirements	Clause	Observations																																													
12.4	Fire prevention in BTPN wagon: Implementation of RDSO guidelines circulated vide letter no MW/BTPN dtd 22.06.12 for fire prevention in BTPN wagon	RB letter no. 2005/M(N)/951/41 dtd 26.06.12																																														
12.5	Inspection of depot by Officers / Supervisors ➤ Inspection of freight depot (sick-line, Yard and ROH depot) is carried out as per RB guidelines: <table><tr><td>Type of inspection</td><td>JA</td><td>SS</td><td>JS</td><td>Sup</td></tr><tr><td>Sick-line</td><td>1</td><td>2</td><td>2</td><td>2</td></tr><tr><td>Yard</td><td>2</td><td>2</td><td>3</td><td>3</td></tr><tr><td>ROH depot</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Super check of goods rake</td><td>1</td><td>2</td><td>4</td><td>4</td></tr><tr><td>Night inspection</td><td>1</td><td>1</td><td>1</td><td>2</td></tr><tr><td>Office inspection</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Brake Power Check (Goods Train)</td><td></td><td></td><td></td><td></td></tr><tr><td>Qualities of GDR check</td><td>1</td><td>1</td><td>2</td><td>4</td></tr></table> ➤ Items during freight examination as per above schedule are checked as per guidelines laid down vide RB letter no. 2009 / M(safety)/7/2/Checklist – inspection dtd 9.09.09	Type of inspection	JA	SS	JS	Sup	Sick-line	1	2	2	2	Yard	2	2	3	3	ROH depot	1	1	1	1	Super check of goods rake	1	2	4	4	Night inspection	1	1	1	2	Office inspection	1	1	1	1	Brake Power Check (Goods Train)					Qualities of GDR check	1	1	2	4	RB letter no. 2009 / M(safety)/7/2/C checklist – inspection dtd 9.09.09	
Type of inspection	JA	SS	JS	Sup																																												
Sick-line	1	2	2	2																																												
Yard	2	2	3	3																																												
ROH depot	1	1	1	1																																												
Super check of goods rake	1	2	4	4																																												
Night inspection	1	1	1	2																																												
Office inspection	1	1	1	1																																												
Brake Power Check (Goods Train)																																																
Qualities of GDR check	1	1	2	4																																												

Annexure-I**CHECKLIST FOR FREIGHT STOCK EXAMINATION POINT**

(Ref: CAMTECH guidelines for Standardization of Infrastructure facilities for freight stock maintenance)

Railway : Type of stock : Type of attention :
 Division: Avg. no. of rakes examined :

SN	Item	Availability		Remarks
		Yes	No	
1	7.5 m center to center distance between two lines			
2	Concrete path way			
3	Paved road for truck movement			
4	Supervisors Room			
5	Staff room			
6	Rolling in/out huts			
7	Field store + Tool room			
8	Compressor Room			
9	Exhauster room			
10	Battery Charging room			
11	Tower light			
12	Street light along examination lines at two levels			
13	Portable rechargeable light			
14	Hand lamps			
15	Welding line			
16	Arc welding set (Mounted on trolley)			
17	Portable DG set			
18	DG set 250 KVA (for compres3ssor, light and welding)			
19	Electric Compressor (500 CFM)			
20	Diesel Compressor (500 CFM)			
21	Exhauster			
22	Gas cutting set			
23	Hyd. Cold cutting set/Hand shears			
24	Protective clothing with Helmet & Cloves			
25	Welding apron & welding glass			
26	Pneumatic pipe line (32 mm dia)			
27	Rake test rig			
28	Hydraulic jacks			
29	Wheel barrow			
30	Pick up van			
31	Truck			
32	Lister truck			
33	Road mobile crane			
34	Hand trolley			
35	Portable drill machine 12 mm			
36	Drill machine			
37	Bending machine			
38	Fitter tool box			
39	Pressure gauge test stand			
40	Folding type portable wheel diameter gauge			
41	Rail profile gauge cum recorder in box			
42	Wheel distance measuring gauge			
43	Buffer height gauge			
44	Tyre defect gauge			
45	Electronic device/Infra red thermometer for detecting warm box			
46	Electric operated hand tools, i.e., Grinder			
47	Pocket jack (Hydraulic) 6 tons capacity			
48	Hand tools			

SN	Item	Availability		Remarks
		Yes	No	
49	Storage rack with pigeon holes			
50	Rack with locker			
51	Plain paper fax with P&T line			
52	Walkie Talkie set/Mobile phone			
53	Railway Telephone			
54	BSNL Telephone			
55	Computer with printer, Scanner, UPS with preloaded software & modem			
56	C&W Office furniture			
57	Railnet / Internet connection			
58	Plain paper copier			
59	First aid equipment			

Table II

**Standard Infrastructural facilities
including M&P for Sick-line Maintenance of freight Stock**

(Ref: CAMTECH guidelines for Standardization of Infrastructure facilities for freight stock maintenance)

SN	Description	Major (>250FWU per month)	Minor (up to 250 FWU per month)	Observations
A	INFRASTRUCTURE			
1	Covered shed Height of crane should be about 10m so that wagon can be lifted by EOT crane and moved over standing wagon to place anywhere in the bay.	100mX75mX12m high Details of layout are shown in sketch	100mX75mX12m high Details of layout are shown in sketch	
2	SAB repair room	5x16m	5 x16m	
3	Air brake & vacuum brake rooms	5x20m		
4	Service building (electrical rooms, staff room on both sides)	5x24m	5x12m	
5	Computer room	5x8m	5x7m	
6	Store room + sub store + tool room	5x26m	5x15m	
7	Machine shop	5x20m	5x8m	
8	Battery room for charging lister/jumbo truck	5x7m	5x3m	
9	Compressor room+ Exhauster room	5x12m	5x9m	
10	Office & SSE/SE/JE rooms	5x16m	5x5m	
11	Sections for attention to various items as detailed in layout	5x44m	5x18m	
12	Room for DG set	At suitable location removed from main shed too contd. noise interference		
13	Total lines required	Six lines under covered shed and one line each out of covered shed for wheels & turn around.	Two lines under covered shed and one line each out of covered shed for wheel and turn around.	
14	Pit line	One pit 20m long (at one end only) on each line depth 960mm from rail level with adequate drainage.		
15	Distance between two lines	10 meters		
16	Pathway between the lines	Heavy duty concrete		
B	LIGHTING			
1	Tower light	Six tower lights per sick line with adequate height	Four tower lights per sick line with adequate height	
2	Street light along examination lines	One row per examination line covering both sides having proper illumination specially covering bogies and brake gear locations. The lights should be at two levels (lower lever for under gear and upper lever for body)		
3	Lighting in the shed and sections	Refer to Annexure III on illumination		
C	MACHINERY & PLANT			
1	EOT cranes 15T/5t	4	2	
2	EOT crane 10T/3T	2	-	
3	Road crane 10T capacity	1	1	
4	Jib crane 2.5T	2	2	
5	Electrical stationary screw air compressor complete with air receiver 15m3/Min (500CFM) required for testing of air brake	2	1	

SN	Description	Major (>250FWU per month)	Minor (up to 250 FWU per month)	Observations
	wagons.			
6	Exhaust (required for testing of vacuum brake wagons)			
7	Turn table for wheel sets	As per lay out		
8	Trestles	12 sets	06 sets	
9	Hydraulic pipe bending machine	1	-	
10	Heavy duty shaping machine	1	-	
11	Centre lathe 12 1/2" capacity	1	-	
12	Radial drilling machine	1	1	
13	Centre lathe 6" capacity	1	-	
14	Bench drilling machine	1	1	
15	Heavy duty pedestal grinder double ended	1	-	
16	Shearing machine 6mmx2200mm wide (required to make patches for body work)	1	1	
17	Battery charger (required for charging batteries of fork lifter, truck and other vehicles)	1	1	
18	Weighing machine 500 kg (for scrap disposal)	1	1	
19	DG set (required to run air compressor welding machine and for lighting in case of power failure)	1 no.(500 KVA)	1no.(350 KVA)	
20	Gas cutting equipment	8	4	
21	Work station for bogie repair	10	1	
22	Fixture for riveting spring plank	2	-	
23	Welding machines	10	6	
24	Winches	3	1	
25	Bogie manipulators	3	-	
26	Pipe line for air 32 mm dia	As per layout		
27	CO2 welding machine	2	-	
28	Rail cum road vehicle	1	-	
29	Fork lifter 3T	4	1	
30	Truck 10T capacity	2	1	
31	Tractor with hydraulic lifting machine & 3 trolleys (trailers)	1	-	
32	Platform, truck 2T	2	1	
33	Light store vehicle	1	1	
34	Hand trolleys with rubber wheels	10	6	
D	TOOLS & EQUIPMENT			
1	Portable hydraulic riveter	1	1	
2	Store bin	20	15	
3	Portable electric grinder	3	2	
4	CBC & draft gear replacement	1	-	
5	SAB test bench	1	-	
6	Hydraulic jacks 15 t	8		
7	Set of special spring spanners. Box spanners. Torque spanners	4sets	2sets	
8	Bottle jacks	8		
9	Pneumatic/electrical hand tools	As per requirement		
10	Gauges & instruments	As per lay out		
11	single wagon test rig	1 no. per 8 wagons per day		
12	DV test stand	2	1	
13	Hand tools	As per requirement		
14	Pneumatic torque wrench with sockets 1" to 2"	4 sets	2sets	

SN	Description	Major (>250FWU per month)	Minor (up to 250 FWU per month)	Observations
15	Ultrasonic leak detector for tank wagons	1	1	
16	Pressure & vacuum gauge dead weight tester	2	1	
E	MISCLLEANEOUS			
1	Water cooler	3	1	
2	Floor cleaning machine	2	1	
3	Fitter Lockers	As per requirement		
4	Fax with P&T line	1	-	
5	Fast aid equipment	4sets	2sets	
6	Photo copier	1	1	
7	VHF sets mobile phone	10sets	6sets	
8	Storage racks	As per requirement		

**Quality audit check-sheet with regard to maintenance of
Stainless Steel Wagon (BOXNHL / BCNHL/BRN22.9)**

(Based on Chapter 3 of CAMTECH Maintenance Manual for Wagon
and CAMTECH Guidelines for maintenance of Stainless Steel Wagons circulated vide
CAMTECH.M/W/SS/Wagon – 1.0 Dec'09)

SN	Requirements					Clause	Remarks
1.0	Facilities required for maintenance of Stainless Steel Wagons:					Para 2.0 of CAMTECH guidelines for maintenance of SS wagons	
	SN	Items	Freight Yard (3 rakes / day)	Sick line (5 Wagons/day)	ROH (200 wagons / Month)		
	1	MIG/MAG welding sets (400 Amp)	02 Nos	02 Nos	03 Nos		
	2	DC Arc Welding Rectifier (600 Amp)	02 Nos	02 Nos	03 Nos		
	3	Portable Air Plasma cutting machine (max capacity 16 mm)	02 Nos	02 Nos	03 Nos		
	4	Consumables for air plasma M/s	As per requirement				
	5	Welding electrodes (Class D & Class M2)	As per requirement				
	6	Huck Bolts of various sizes	As per requirement				
	7	Lock bolts equipment	2 sets	2 sets	3 Sets		
	8	Portable grinding machine)	03 Nos	03 nso	05 Nos		
	9	SS wire brush	As per requirement				
	10	SS (IRS M44 plates)	As per requirement				
2.0	Equipments for Lock Bolt installation, air plasma cutting machine, DC arc welding rectifier should be procured as per Specification laid down in CAMTECH guidelines					Para 3 of CAMTECH guidelines for maintenance of SS wagons	
3.0	Welders are qualified for welding of SS to SS and SS to Mild steel					-do-	
4.0	Qualified personnel are available to carry out lock bolting,					-do-	
5.0	Sick line has facility for EOT cranes with covered area					Ch 3 of CAMTECH Manual for	

SN	Requirements	Clause	Remarks
		Wagon	
6.0	Air brake testing facility: iii) Compressor of adequate capacity to carry out air brake examinations of rake during intensive examination as well as air brake testing of wagon in sick line . Pneumatic line is provided with moisture trap or air-dryer to ensure availability of moisture free air for air brake testing. iv) Availability of adequate RTR for air brake testing during intensive examination and SWTR for air brake testing in sick line.	Ch 3 of CAMTECH Manual for Wagon	
7.0	Infrastructure facility for Yard maintenance viii) Adequate centre to centre distance between tracks (at least 7.5 mts) for conducting intensive examination. ix) Concrete pathway from one end to other end. Material handling facility to ensure movement of material from one end to other end. x) Proper illumination for conducting intensive examination. xi) Welding facility (Rectifier type) for welding during intensive examination. xii) Adequate outlets for tapping air pressure for testing of stock. xiii) Duty room for maintenance supervisor, compressor room, store room, tool room, welding machine room for intensive examination. xiv) VHF sets for close monitoring. Infrastructure facilities for Yard attention to be checked as per Annexure – I of CAMTECH report for infrastructure facilities	Ch 3 of CAMTECH Manual for Wagon	
8.0	Other facilities and M&P's: i. Wheel diameter measuring gauge. ii. Wheel gauge. iii. Transportation facility for wheel sets and CTRBs. iv. Availability of general tools for maintenance (Wrenches, hammer, chisel, anvil etc). v. Wheel defect gauge. vi. Gauges for measuring 'A" dimension & piston stroke. vii. Electrodes heating chamber.	Ch 3 of CAMTECH Manual for Wagon	
9.0	Analysis of sick marking and yard attention is being carried out. Computer facility for such analysis is available.	-	
10.0	Supervisors and staffs are trained with regard to following: ➤ Air brake testing of conventional BOXNHL / BCNHL stock as well as BOXNHL –MBS stock ➤ Construction details of BOXNHL / BCNHL for yard examination and attention as well as sick line attention. ➤ Huck bolting process for attention at Yard / Sick line. ➤ MIG / MAG welding process / MMAW welding process for SS welding. ➤ Plasma cutting of SS sheets. ➤ Attention of BOXNHL / BCNHL door	-	
11.0	Procurement of stores for maintenance of SS wagon (BOXNHL/BCNHL) i) List of stock items. ii) List of non stock items. iii) Storage facility with covered area for critical stores. iv) Maintenance of ledger and consumption for last two years and current year. Spares to be procured as stock items include items required for regular consumption of rolling stock.		

SN	Requirements	Clause	Remarks
12.0	General housekeeping:		
	Level of Housekeeping		
	Overall ambience and working condition		
	Storage of Wheel sets		
	Storage of Air brake cylinder, BMBS, AR and other stores.		

Depot details

Depot name:

Division and Railway:

Activities carried out:

SN	Description	Observations
1	ROH outturn per month (Av of last one year in VU)	
2.	Nos of CC rake running in service with brake up of formation (Rake type wise)	
3	Average examination per day (CC/ Premium/ end to end)	
4	Average yard detachment per rake (Av of last one year)	
4.1	In CC rake	
4.2	In premium rake	
4.3	In end to end examined rake	
5.	Sick line outturn per month (Av of last one in VU)	
6	Training of welders for MMAW / MIG welding	
7	Training of staff in Oxy cutting	
8	Training of staff in Plasma cutting	
9	Availability of critical and safety stores	
10	Whether a dedicated store officer is nominated for urgent procurement	
11	Sanctioned imprest value of depot officer	
12	Whether depot is ISO 9001 certified	
13	Availability and display of work instruction in ROH depot /sick-line / yard	
14	Overdue refresher course of technician	
15	Record keeping: a) ROH attention - Causes of replacement/rejection of components replaced/rejected during ROH are to be recorded in a separate register. Assessment of monthly consumption and trend of failure can be done by analyzing these records. b) Record for Yard attention with regard to manpower deployment, damage deficiency register, consumption of stores etc. c) Record for sick-line attention stock-wise with man-hour utilization and consumption of stores	